Cipherops

PQC DNSSEC MTL Benchmark Using a Reference Implementation (PS-036)

How it works: architecture, pipeline, measurements and results

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Agenda

From the problem to the numbers, in the order a run executes.

1. **The problem**: post-quantum signatures in DNS
2. **MTL mode** in one picture
3. **What PS-036 asks** and how we measure it
4. **Architecture**: components, container, data flow
5. **How to run it**
6. **The 7 stages of a run**, with real inputs and outputs
7. **Results**: signing, verification, answer sizes, bytes on the wire, cache
8. **The result in one table**
9. **What is measured vs modelled**, limitations
10. **Reproduce and learn**: commands, trace script, docs

Post-quantum signatures do not fit today's DNS

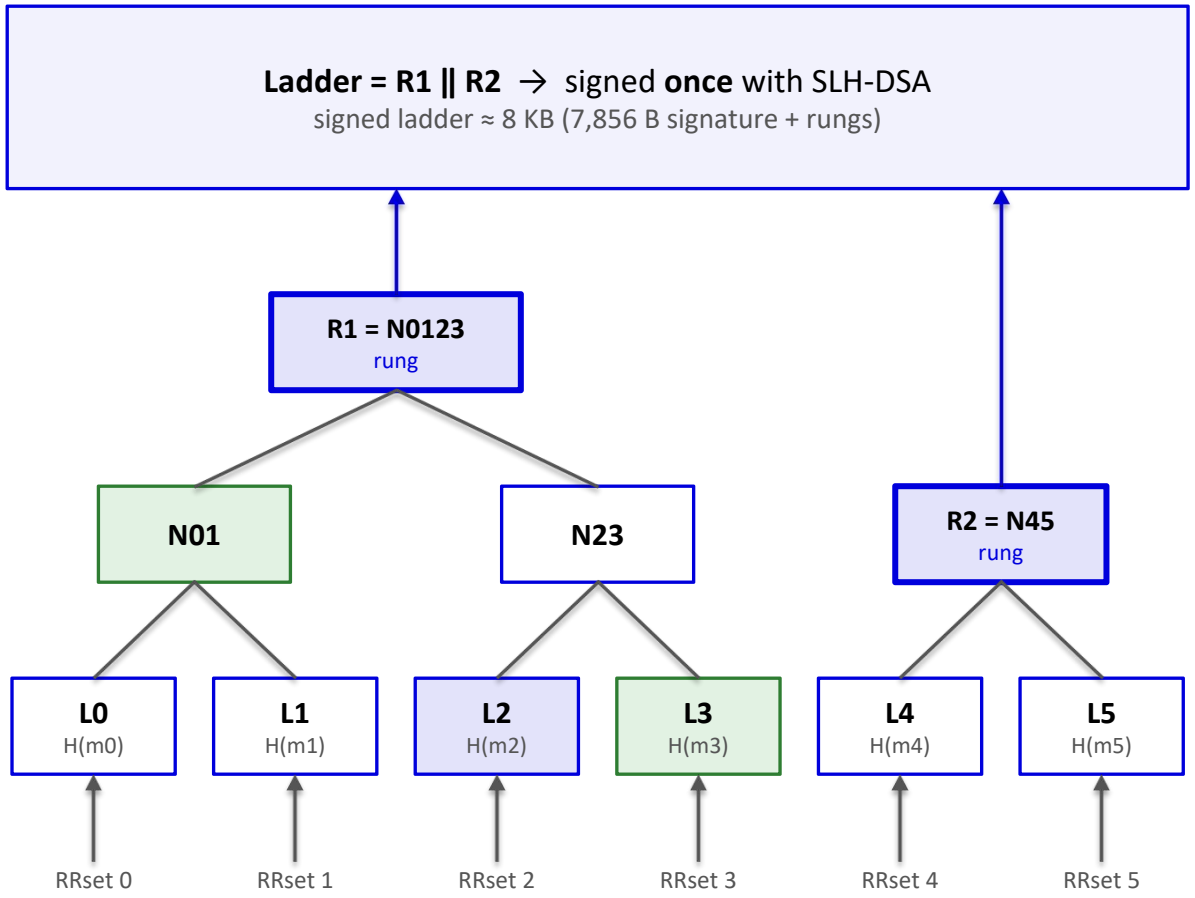
Every DNSSEC answer carries a signature. Post-quantum signatures are huge and slow to make.

	Ed25519 / ECDSA	RSA-2048	SLH-DSA-SHA2-128s (post-quantum)
Signature in every answer	64 B	256 B	**7,856 B**
Sign one record set	≈ 30 µs	≈ 0.6 ms	**≈ 0.5 s**
Small answer (one IP address)	178 B	370 B	**7,970 B**
Fits a 1,232-byte UDP answer?	yes	yes	**never → truncated, retried over TCP**

- Re-signing a 10,005-record-set zone with SLH-DSA takes about **80 minutes** on one core.
- **MTL mode** (Merkle Tree Ladder) promises to fix size and speed: sign the whole zone **once**, give each answer a small hash path. **PS-036**: measure whether it really does, with the reference implementation.

MTL mode in one picture

Example zone with 6 record sets. Real zones: 45, 125, 405 (quick demo) up to 10,005.



6 = binary 110 → 2 rungs (one per 1-bit). 45 = 101101 → 4 rungs.

Key terms

- **Condensed signature** of RRset 2 = the hash path to its rung: **L3, N01** (green). That, plus a 1-byte type, is what goes in its RRSIG: **156–204 B** in our zones instead of 7,856 B.
- **Signed ladder** = the rungs + one SLH-DSA signature. One per zone, about **8 KB**. A resolver fetches it once (EDNS option mtl-mode-full, over TCP) and caches it.
- **Full signature** = condensed + signed ladder. Used only when the resolver has no ladder yet.
- Verify with a cached ladder: recompute L2 → N23 → R1, compare with the ladder: **hashes only**.

What PS-036 asks, and how we measure it

Every requirement maps to one stage of the run and one output file.

PS-036 requirement	How Cipherops does it	Where the number is
Use the supplied reference implementation	Verisign MTL library, unmodified, pinned commit 7cef019	environment.json
Defer if it or its fixtures fail	Go/no-go gate: Verisign test suite, tool round trip, harness smoke test; NO-GO stops the run (exit 3)	preflight.json
One PQC algorithm, with and without MTL	SLH-DSA-SHA2-128s in MTL mode vs signing every RRset; Ed25519, ECDSA, RSA as references	trials.csv
Signing time, verification time	C harness times every library call in nanoseconds; warm and cold verification	trials.csv, ops.parquet
Transmitted bytes	Real DNS answers with the real signatures; UDP, truncation, TCP; 301-lookup stream	responses.csv, workload.csv
Cache usage	Resolver cache measured (dnspython LRUCache); draft §12 formulas alongside	cache.csv, sizes.csv
Several dataset sizes, repeated trials	3 zone sizes; 3 trials (5 in the full run); medians with 95% bootstrap CI	summary.json, report

Solution Architecture & Design

UML component view: one pinned container runs every stage and calls the unmodified reference libraries.

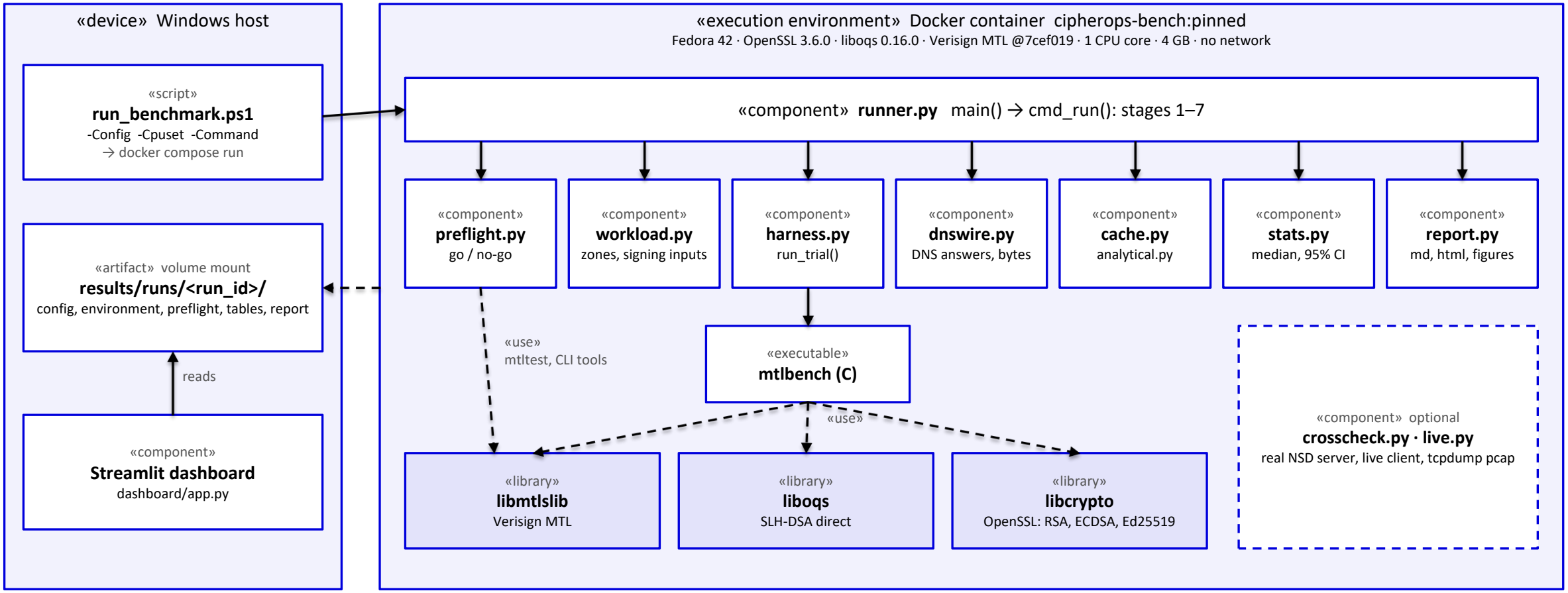


Figure: System Architecture UML Diagram

Data Flow Diagram

Level-1 DFD of one run: from the configuration to the report.

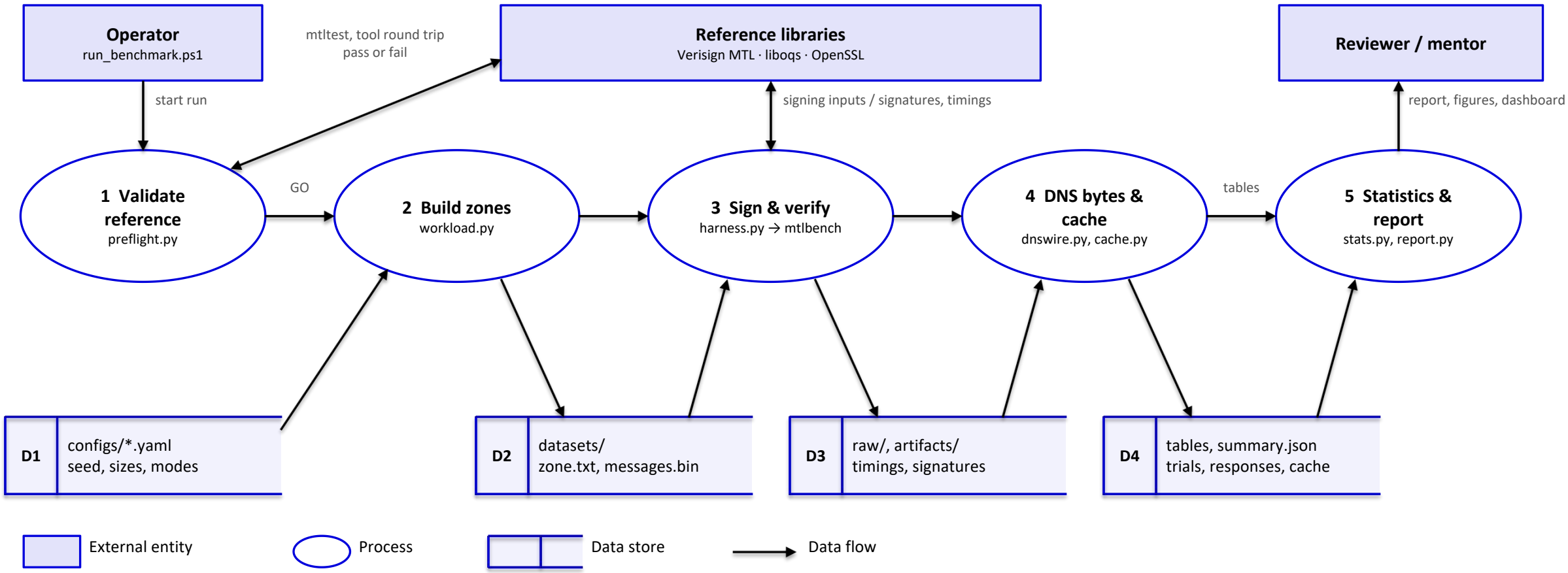


Figure: Data Flow Diagram (DFD)

Pinned environment and how to run it

One image, one command. The host needs only Docker and git.

Component	Version (pinned)
Base image	Fedora 42 (fedora-minimal)
OpenSSL	3.6.0
liboqs	0.16.0 (Release)
Verisign MTL	commit 7cef019, -O2
Harness	mtlbench (C), built in the image
Python	3.13, dnspython 2.8, pandas, numpy
Limits	1 CPU core (cpuset 2), 4 GB, no network

Commands (Windows PowerShell, repository root)

Full PS-036 run (50 / 500 / 5,000 names, 5 trials)

```
.\scripts\run_benchmark.ps1
```

Quick demo (20 / 60 / 200 names, 3 trials)

```
.\scripts\run_benchmark.ps1 -Config configs/quick.yaml
```

Go/no-go gate only

```
.\scripts\run_benchmark.ps1 -Command preflight
```

Dashboard

```
streamlit run dashboard/app.py
```

The script builds the image, records git SHA and image ID, then runs python3 -m cipherops.runner run inside the container. Results appear in results\runs\<run_id>\.

One run = 7 stages

runner.py cmd_run() executes them in order. Values from the quick demo run.

Stage	Code	Input	Output
1 Config + environment	load_config, envinfo	quick.yaml: seed 20261005, 3 trials, 5 modes, 20 / 60 / 200 names	config.yaml, environment.json
2 Go/no-go gate	preflight.py	Verisign library, tests, tools; mtlbench	preflight.json: GO (9 checks)
3 Zones + signing inputs	workload.py	names, seed, modes	45 / 125 / 405 RRsets; 15 × messages.bin
4 Sign + verify	harness.py → mtlbench	15 messages.bin files	45 trials: trials.csv, raw/, artifacts/ (signatures)
5 DNS bytes + cache	dnswire.py, cache.py	zones, real signatures, 301 lookups	responses, workload, cache, sizes .csv
6 Statistics	stats.py, summarise	45 trial rows + stage 5 tables	summary.json (median, 95% CI)
7 Report	report.py	the run folder	report.md / .html, 5 figures

Stages 1–2: configuration, environment, go/no-go

Before any number is produced we record what we run on, and prove the reference library works.

Stage 1 – inputs and record

- **configs/quick.yaml**: seed 20261005, 3 trials + 1 warm-up, 5 modes, zones of 20 / 60 / 200 names, 16 cold MTL verifications per trial, direct SLH-DSA capped at 60 / 100 signatures, 300 lookups.
- **Run folder**: results/runs/**20261007T045213Z-quick-demo**/
- **environment.json**: Intel Core Ultra 7 155U, pinned core [2], WSL2 kernel, OpenSSL 3.6.0, liboqs 0.16.0, MTL 7cef019 (-O2), git SHA, image ID, SHA-256 of mtlbench.

Stage 2 check	Result
Library files installed	ok
Verisign test suite (mtltest)	89 passed, 0 failed
mtlkeygen → mtlsign → mtlverify, 16 msgs + 1 tampered	all verified, tamper rejected
mtlbench smoke test, 6-name zone, 5 modes	all pass, tamper rejected
Bundled vector scheme (warning only)	ML-DSA-44-MTL
Verdict	**GO**

NO-GO on any critical check → the run prints the verdict and stops with exit code 3. No benchmark numbers are produced (PS-036: defer).

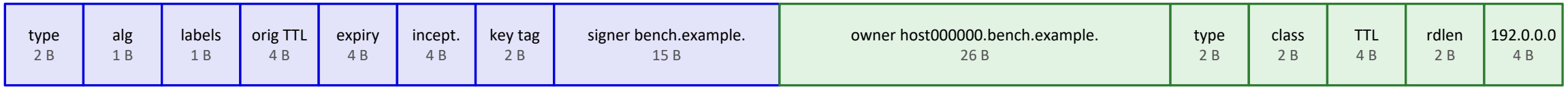
Stage 3: zones and signing inputs

Deterministic DNS zones, then the exact bytes each signature must cover (RFC 4034 §3.1.8.1).

Zone	Names	RRsets	apex	small	medium	large	NSEC3
small	20	45	4	14	3	3	21
medium	60	125	4	32	17	11	61
large	200	405	4	117	70	13	201

- Zone **bench.example.**, seed + dataset index
- **small** 60%: one A / AAAA · **medium** 30%: 3 MX or 4 AAAA · **large** 10%: TXT 1–2 KB
- RRsets = 2 × names + 5 (4 apex, 1 per name, NSEC3 chain)

One signing input (here host000000 A, 73 bytes)



RRSIG fields, signature left empty: 33 B

the RRset in canonical form: 40 B

- Per mode only the algorithm byte, key tag and DNSKEY change, so each of the **15** (zone, mode) pairs gets its own **messages.bin**: small/MTL = 45 messages, 9,311 B. Keys do not exist yet: DNSKEY holds a placeholder of the right length.
- Direct SLH-DSA is capped at 60 / 100 evenly spaced RRsets (**messages_sample.bin**); its zone total is extrapolated.

Stage 4: sign and verify (one trial)

45 trials = 3 trials × 3 zones × 5 modes, shuffled order, one new process and new key pair each.



MTL trial, small zone (45 RRsets)	Calls	Time
Key generation	1	68.6 ms
Add each RRset as a leaf (hash)	45	3.9 µs each
Sign the ladder (SLH-DSA)	**1**	**486 ms**
Extract condensed signatures	45	0.24 µs each
Verify the ladder	1	441 µs
Verify an answer, ladder cached	45	**2.4 µs each**
Verify a full signature (cold)	16	604 µs each
Tamper: message, signature, ladder	3	all rejected

Other modes, same zone

- **SLH-DSA direct**: 45 × sign ≈ **0.50 s**, 45 × verify ≈ 0.47 ms.
- **Ed25519**: 45 × sign ≈ 31 µs, verify ≈ 93 µs. ECDSA signatures converted to DNSSEC's 64-byte form.
- **Outputs**: raw/*.json.gz (45, exact JSON), artifacts/ (trial 0 signatures, used in stage 5), trials.csv (45 rows), ops.parquet (16,899 timed calls).
- All 45 trials: exit 0, 0 failures, every tamper test rejected.

Stage 5: DNS answers, bytes on the wire, cache

Real DNS messages (dnspython) with the real signatures inside. No verification here: sizes and traffic.

host000000 A, buffer 1,232 B	Answer	Signature	Transport	Bytes both ways
Ed25519	178 B	64 B	UDP	231
SLH-DSA + MTL (ladder cached)	271 B	157 B	UDP	324
SLH-DSA + MTL, first contact (full)	8,295 B	8,181 B	TCP	8,356
SLH-DSA direct	7,970 B	7,856 B	UDP → TC=1 → TCP	8,133

Transport model (dnswire.exchange)
fits: query + answer
too big: query + TC reply + (2 + query) + (2 + answer)
MTL full: (2 + query) + (2 + answer) over TCP

- **responses.csv**: every answer measured once (DNSKEY, each name, 50 NXDOMAIN) at 1,232 and 4,096 B.
- **cache.csv**: answers loaded into dnspython's cache and measured.

The lookup stream (workload.csv)

- **301 lookups**: DNSKEY first, then 300; 10% for names that do not exist.
- Popularity follows **Zipf (s = 1.1)**: rank r gets weight 1/(r+1)^1.1, like real DNS traffic.
- A repeat lookup is served from the **resolver cache**: only **48 / 74 / 106** answers are sent.
- MTL: after the first answer, **one** mtl-mode-full fetch of the ladder over TCP.

Stages 6–7: statistics and report

Medians over trials with a 95% bootstrap confidence interval; one report per run.

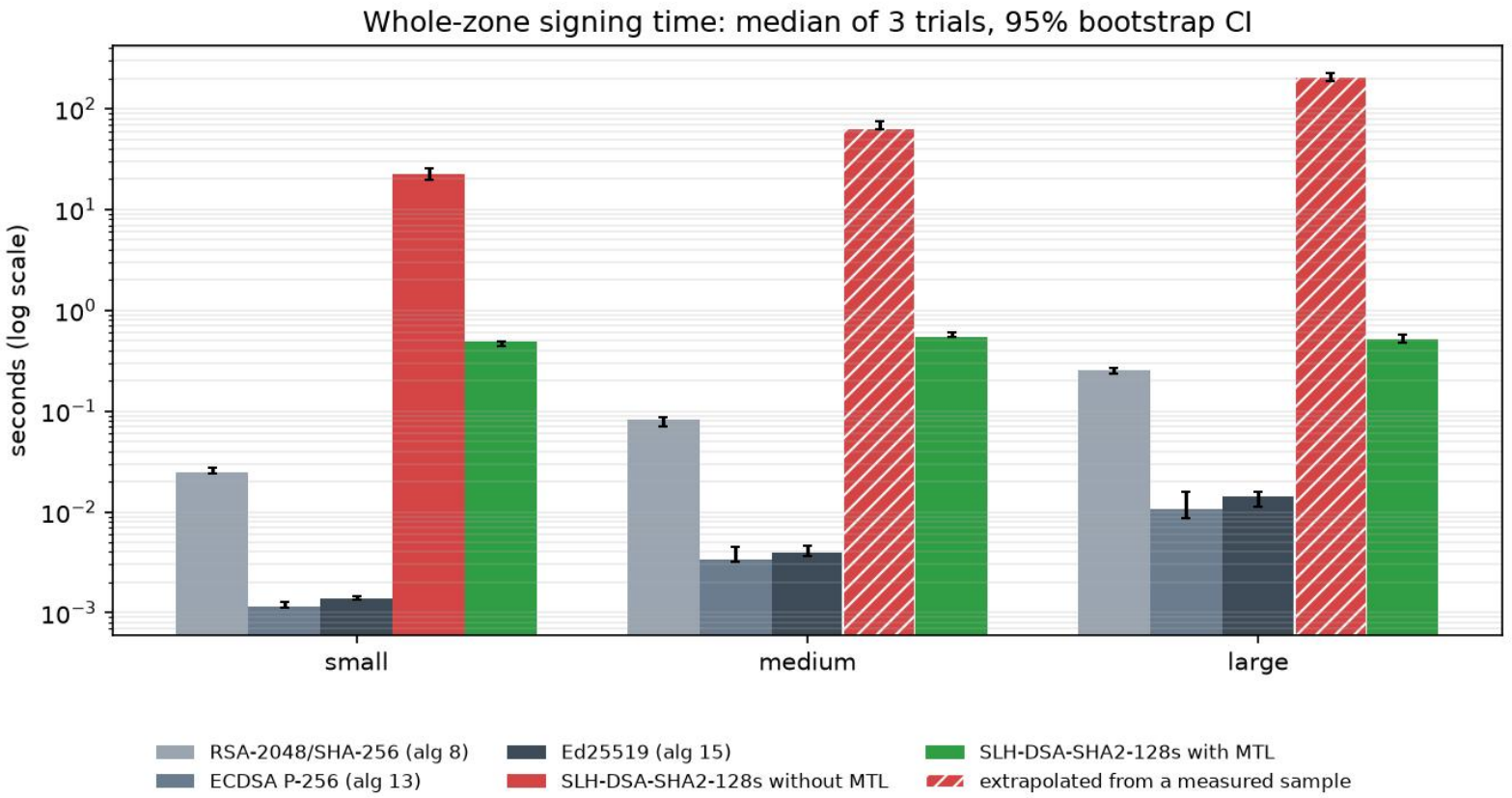
- **Input**: 45 trial rows (3 values per zone × mode) and the stage 5 tables.
- **stats.summarize**: n, mean, **median**, std, p05, p95, min, max.
- **stats.bootstrap_ci**: resample the 3 trial values 2,000 times, take each median, keep the 2.5th–97.5th percentile.
- **summary.json**: trials, transmitted bytes, UDP fit per record class. Example: large MTL signing median **0.516 s**, CI [0.472 s, 0.575 s].
- **all_checks_passed** combines exit codes, failures and tamper tests per zone × mode.

Stage 7 – report.write(run)

- **report.md / report.html**: problem, method, all tables, discussion, limitations, appendices with config, environment and gate.
- **figures/**: sign, verify, responses, workload, cache (next slides).
- **Streamlit dashboard** reads the same run folder.
- Exit code: 0 all checks passed · 1 a trial failed · 3 NO-GO.

Signing time: MTL signs the zone once

Whole-zone signing time per mode and zone size.



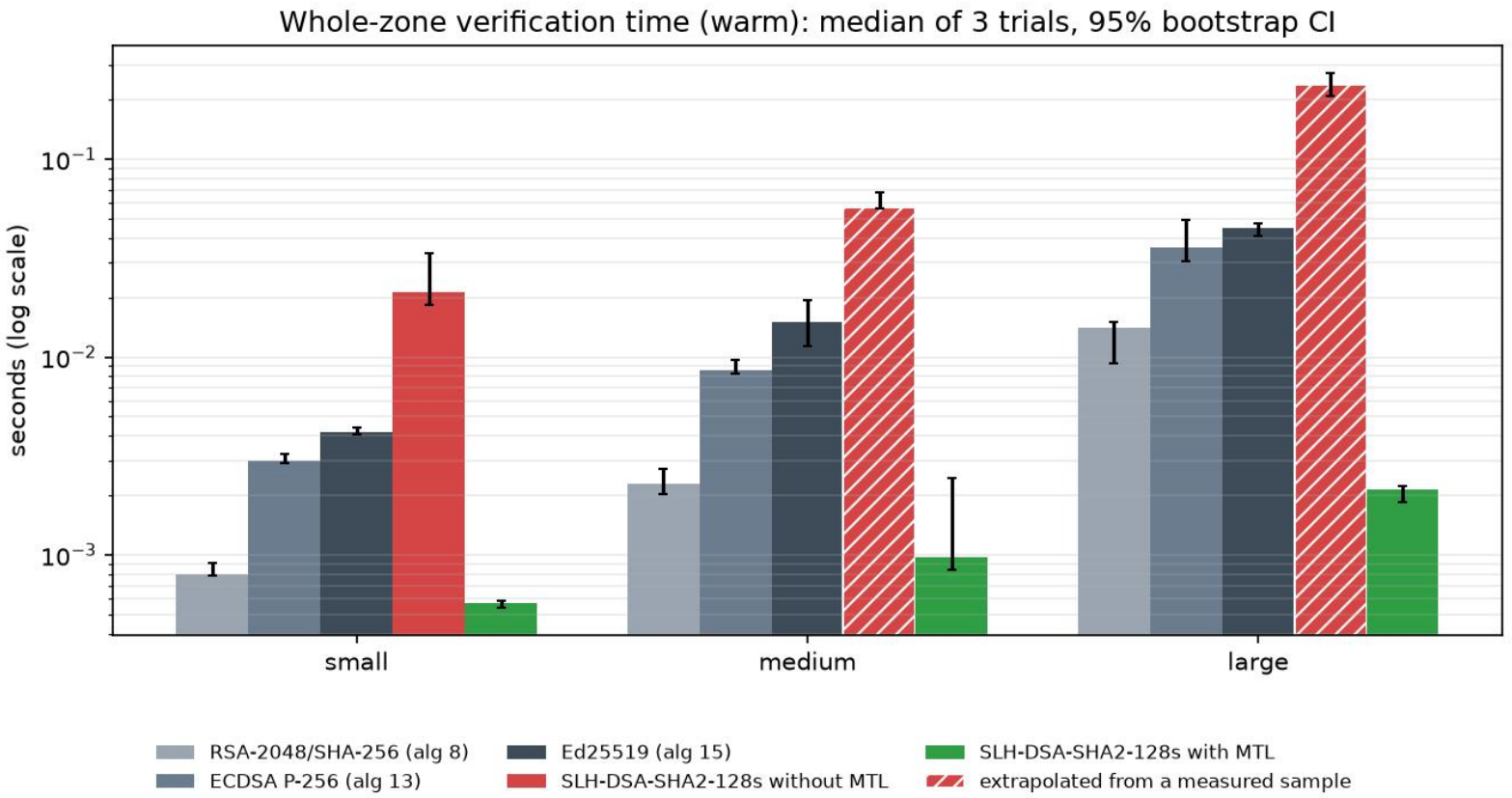
Large zone (405 RRsets)

- SLH-DSA direct: **≈ 209 s** (100 signed at 0.52 s, × 405)
- SLH-DSA + MTL: **0.52 s** – one ladder signature + 405 hashes
- Ed25519 0.014 s · ECDSA 0.011 s · RSA 0.25 s
- MTL stays ≈ 0.5 s at every zone size: cost is per zone, not per record.

Quick demo run, log scale. Source: trials.csv, median of 3 trials, whiskers = 95% CI, hatched = extrapolated.

Verification time: hashes instead of SLH-DSA

Whole-zone verification with a cached ladder (warm).



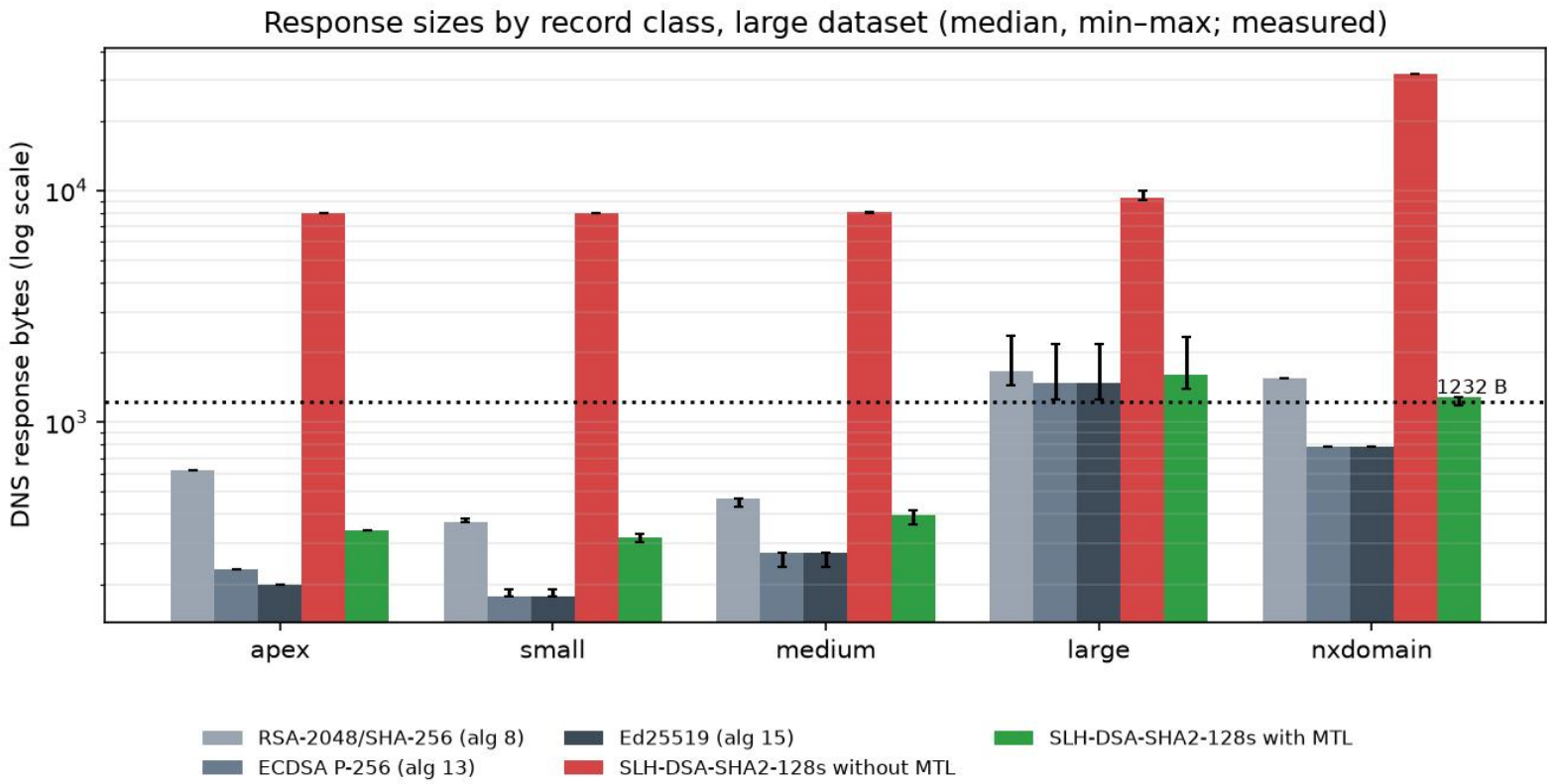
Quick demo run, log scale. Source: trials.csv, median of 3 trials, whiskers = 95% CI.

Per answer, large zone

- Warm: SLH-DSA direct **588 µs** · MTL **5.3 µs** · Ed25519 111 µs
- Cold (first answer, no ladder): direct 511 µs · MTL **513 µs** · Ed25519 103 µs
- MTL pays one SLH-DSA verification per zone (the ladder), then only hashes.

Answer sizes: MTL fits in UDP again

Every DNS answer measured, large zone, by record class.



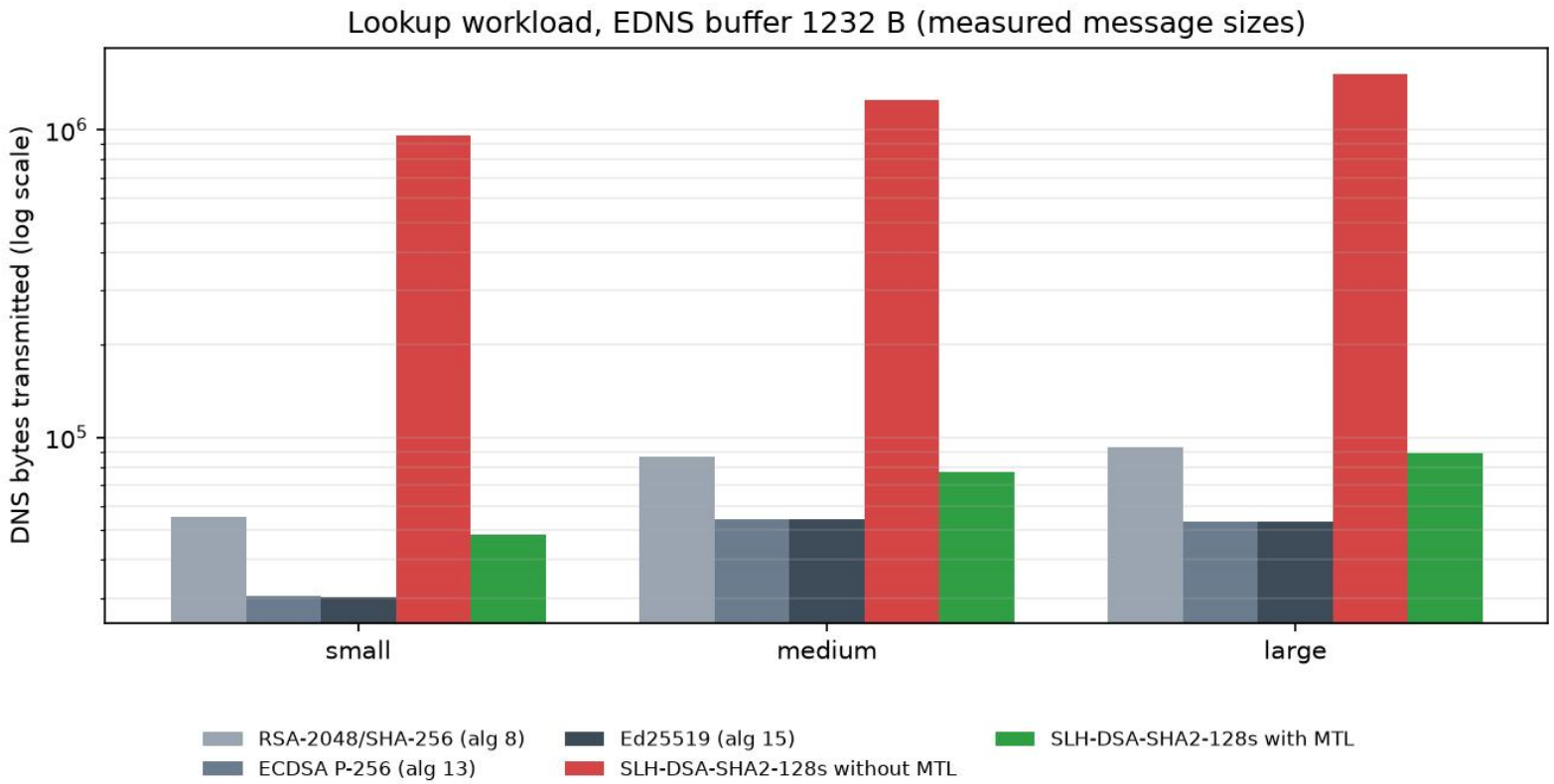
Share of answers ≤ 1,232 B

- Small answers: direct **0%** (7,970 B) · MTL **100%** (319 B) · Ed25519 100% (178 B)
- All answers, large zone: direct 0% · MTL 78% · Ed25519 95%
- Over the line for MTL: large TXT records and some NXDOMAIN (2–4 signatures each).

Quick demo run, log scale. Source: responses.csv, every answer measured once (median, min–max).

Bytes on the wire: 17× less than direct SLH-DSA

Total DNS bytes for the 301-lookup stream at a 1,232-byte buffer.



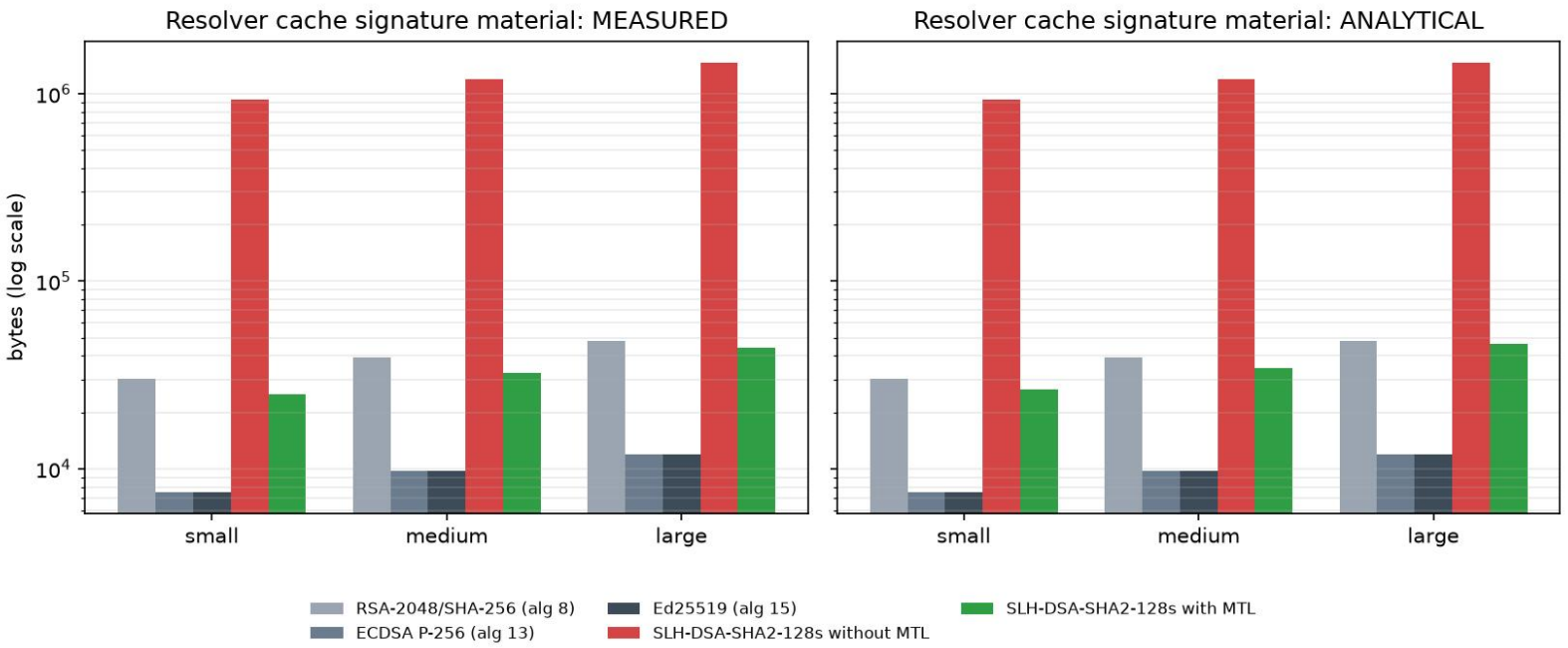
Quick demo run, log scale. Source: workload.csv, one deterministic 301-lookup replay per zone and mode.

Large zone, 106 answers sent

- SLH-DSA direct **1,522,371 B**, 106 TCP retries
- SLH-DSA + MTL **88,895 B**, 33 TCP (32 truncated + 1 ladder fetch)
- Ed25519 53,292 B, 7 TCP · RSA 92,610 B, 34 TCP
- MTL ≈ 1.7× Ed25519 and below RSA-2048.

Cache and storage: 36× less than direct SLH-DSA

Signature bytes a resolver holds after the workload: measured (left), draft formulas (right).



Quick demo run, log scale. Source: cache.csv and sizes.csv.

Large zone

- Resolver cache: direct **1.47 MB** · MTL **44 KB** (incl. 8 KB ladder) · Ed25519 12 KB
- Zone signature storage: direct **3.18 MB** · MTL **0.087 MB** · Ed25519 0.026 MB
- Measured MTL sizes match the draft: condensed 204 B = formula 204 B; ladder 8,056 B ≤ 8,216 B max.

The result in one table

Large quick-demo zone, 405 record sets, medians of 3 trials.

	SLH-DSA direct	SLH-DSA + MTL	Ed25519 (today)
Signature per RRSIG	7,856 B	≤ 204 B + one 8,056 B ladder per zone	64 B
Sign whole zone	≈ 209 s	**0.52 s**	0.014 s
Verify one answer, ladder cached	588 µs	**5.3 µs**	111 µs
Verify first answer, nothing cached	511 µs	513 µs	103 µs
Small answers within 1,232 B	0%	**100%**	100%
Bytes on the wire, 301 lookups	1,522,371 B	**88,895 B** (17× less)	53,292 B
Signature storage for the zone	3.18 MB	**0.087 MB** (36× less)	0.026 MB
All correctness checks	passed	passed	passed

Conclusion: MTL makes SLH-DSA practical for DNSSEC except at first contact, where one ladder fetch and one SLH-DSA verification remain.

What is measured, what is modelled

Every number is labelled so a reviewer knows how much to trust it.

Measured directly	Modelled or estimated
Signing and verification time of every library call (ns, pinned core)	Direct SLH-DSA zone totals on medium / large: sample of 60 / 100 × RRsets
Exact byte size of every DNS query and answer (dnspython encoding)	Transport: UDP / TC / TCP exchange sums; IP fragments and TCP segments by arithmetic
Signature and ladder sizes from the reference library	Resolver behaviour: cache holds everything (TTL 3600), one ladder per zone
Resolver cache contents (dnspython LRUCache, tracemalloc)	Draft §12 maxima shown next to measured

Known limitations

- DNSKEY holds a placeholder key of the right length (keys are made inside each trial), so the zone is size-exact but not validatable by an outside resolver. maxrss reports the Python parent, not mtlbench.
- Quick demo: 3 trials, ≤ 200 names. Full run: 5 trials, 50 / 500 / 5,000 names, plus optional NSD cross-check and packet capture.

Reproduce, inspect, learn

Every number can be regenerated and traced to the line of code that produced it.

Want to…	Use
Run the benchmark	.\scripts\run_benchmark.ps1 [-Config configs/quick.yaml]
See every variable of stages 1–7 on tiny zones	own.py in the container → results\own_trace.txt
Look up any output file or column	docs\outputs-reference.md
Follow inputs → outputs stage by stage	docs\end-to-end-flow.md
Read the full report	results\runs\<run_id>\report.html
Explore interactively	streamlit run dashboard/app.py
Method and design	docs\methodology.md, docs\architecture.md

Thank you

Questions?

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